

Week 2: Analytics in the Age of AI

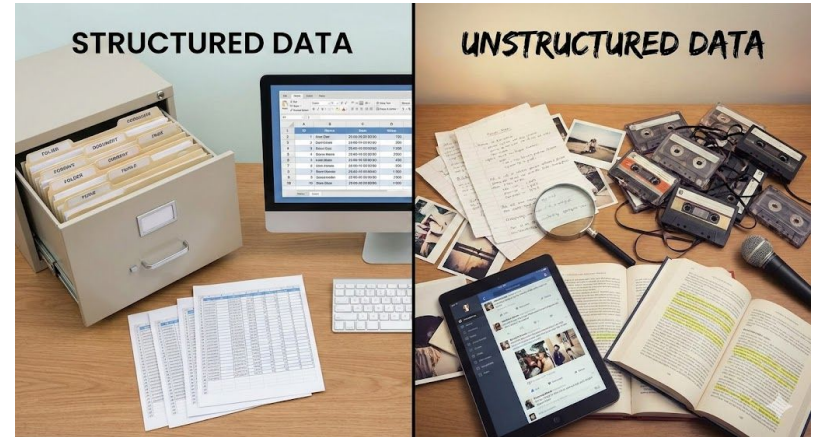
Course: BUSM2725 AI and Analytics for Business Applications

Outline

- Recap of Data in AI/Analytics (structured vs unstructured)
- Activity: Data collection with Google AI Mode
- Activity: Analysing Youtube video content
- Walkthrough: Introducing Google Colab
- Demo: Use Google Colab to analyse Melbourne CBD data
- Use this AI tool to help to gain insights of the case study (Assessment 1) dataset

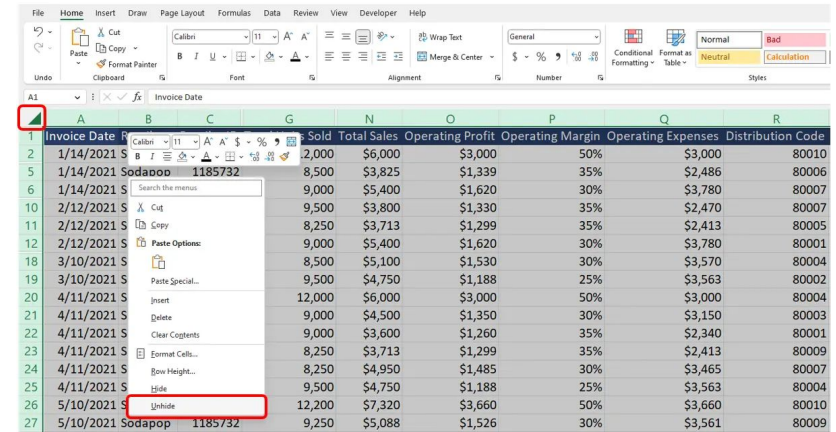
Structured vs Unstructured Data

- What category the following data belongs to?
 - Sales data
 - Customer review
 - Voice recording
 - Registration form
 - Signal from roadside units
 - Readings from smart meters
 - HR performance
 - Inventory records
 - Footage from security camera



Structured Data

- **Definition:** Highly organized information that fits readily into strict data models, typically stored in relational databases (RDBMS).
- **Characteristics:** It is quantitative, follows a rigid schema, and fits neatly into rows and columns.
- **Examples:** SQL databases, Excel spreadsheets, transaction logs, dates, and phone numbers.



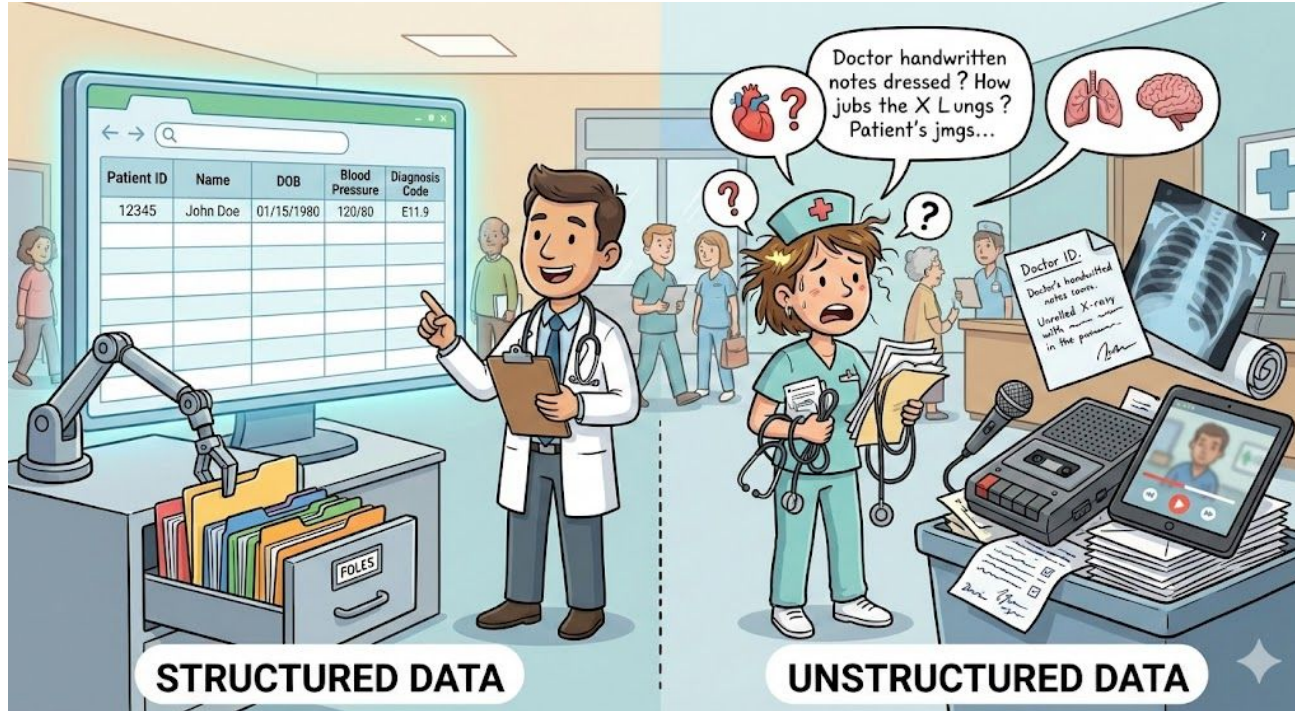
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Invoice Date	Customer			Sold	Total Sales	Operating Profit	Operating Margin	Operating Expenses	Distribution Code								
1	1/14/2021	Sodapop	1185732		2,000	\$6,000	\$3,000	50%	\$3,000	80010								
5	1/14/2021	Sodapop	1185732		8,500	\$3,825	\$1,339	35%	\$2,486	80006								
6	1/14/2021	Sodapop	1185732		9,000	\$5,400	\$1,620	30%	\$3,780	80007								
10	2/12/2021	Sodapop	1185732		9,500	\$3,800	\$1,330	35%	\$2,470	80007								
11	2/12/2021	Sodapop	1185732		8,250	\$3,713	\$1,299	35%	\$2,413	80005								
12	2/12/2021	Sodapop	1185732		9,000	\$5,400	\$1,620	30%	\$3,780	80001								
18	3/10/2021	Sodapop	1185732		8,500	\$5,100	\$1,530	30%	\$3,570	80004								
19	3/10/2021	Sodapop	1185732		9,500	\$4,750	\$1,188	25%	\$3,563	80002								
20	4/11/2021	Sodapop	1185732		12,000	\$6,000	\$3,000	50%	\$3,000	80004								
21	4/11/2021	Sodapop	1185732		9,000	\$4,500	\$1,350	30%	\$3,150	80003								
22	4/11/2021	Sodapop	1185732		9,000	\$3,600	\$1,260	35%	\$2,340	80001								
23	4/11/2021	Sodapop	1185732		8,250	\$3,713	\$1,299	35%	\$2,413	80009								
24	4/11/2021	Sodapop	1185732		8,250	\$4,950	\$1,485	30%	\$3,465	80007								
25	4/11/2021	Sodapop	1185732		9,500	\$4,750	\$1,188	25%	\$3,563	80004								
26	5/10/2021	Sodapop	1185732		12,200	\$7,320	\$3,660	50%	\$3,660	80010								
27	5/10/2021	Sodapop	1185732		9,250	\$5,088	\$1,526	30%	\$3,561	80009								

Unstructured Data

- Information that does not have a pre-defined data model or is not organized in a pre-defined manner. It accounts for 80-90% of all enterprise data.
- **Nature:** Qualitative, massive volume, and difficult to process with traditional tools.
- **Format:** Stored in its native format (Data Lakes).
- **Examples:** Emails, social media posts, videos, audio recordings, PDFs, and images.



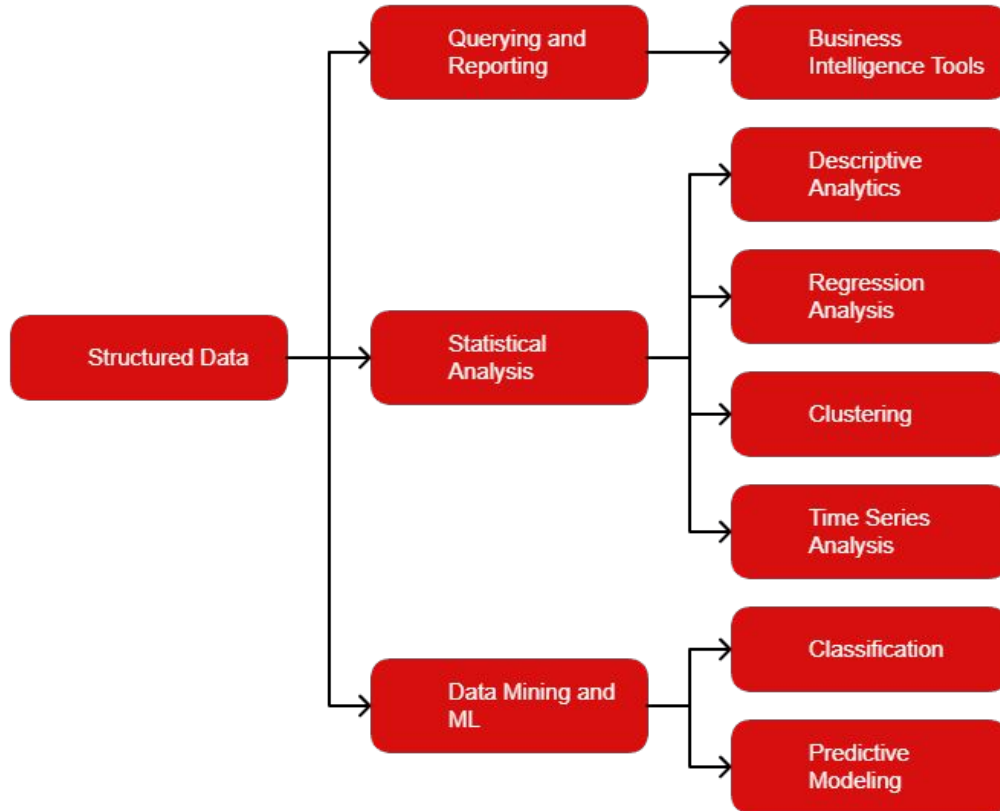
Example: Hospital Data



What are the structured and unstructured data that you might find in a hospital?

How important is it to understand unstructured data in this context?

Standard Ways to Analyze Structured Data



What techniques
did you
experience with?

Activity: Collect data using Google AI Mode

find the results of 10 recent games of Real Madrid and create a table with the following columns: Play againts, Home/Away, Result (Win/Loss), Score, Date ^

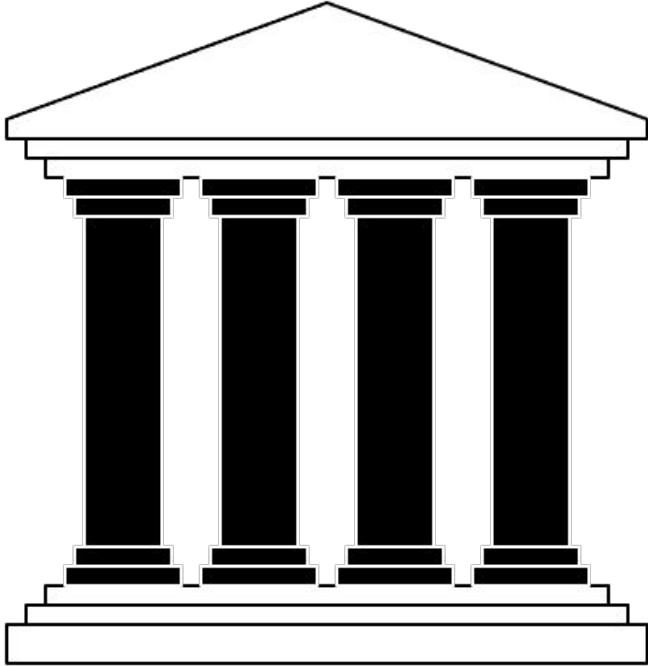
Example: get football stats

The results of Real Madrid's 10 most recent games are provided in the table below, with the data accurate as of November 25, 2025. 

Play Against 	Home/Away	Result (Win/Loss)	Score	Date
Elche	Away	Draw	2-2	Nov 24, 2025
Rayo Vallecano	Away	Draw	0-0	Nov 9, 2025
Liverpool	Away	Loss	1-0 (L)	Nov 4, 2025
Valencia	Home	Win	4-0	Nov 1, 2025
Barcelona	Home	Win	2-1	Oct 26, 2025
Juventus	Home	Win	1-0	Oct 22, 2025
Getafe	Away	Win	0-1 (W)	Oct 19, 2025
Villarreal	Home	Win	3-1	Oct 4, 2025

Collect some data of interest and share with your classmates

Approaches to Analyse Unstructured Data



Natural Language Processing

Techniques for extracting meaning from text data.



Computer Vision

Methods for interpreting and analyzing visual data.



Audio Analysis

Processes for extracting insights from audio files.

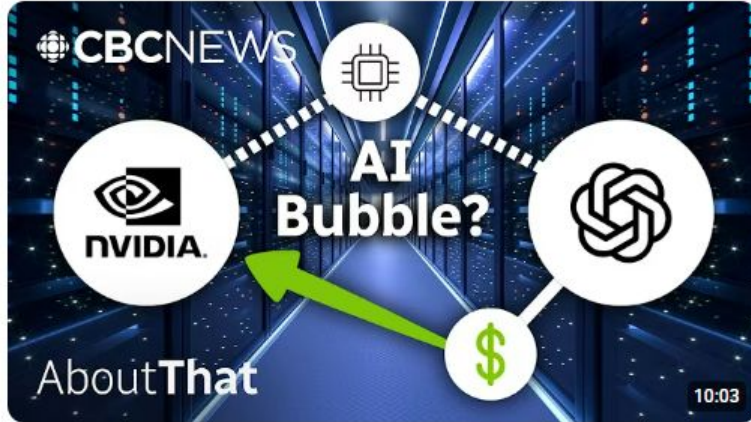


Machine Learning and AI

Algorithms for finding patterns and insights in data.

What techniques did you experience with?

Activity: Analysing Youtube video content



If the AI bubble pops, will the whole U.S. economy go with it? | About That
52K views • 4 hours ago



As investors pour billions into artificial intelligence, warnings of a looming AI bubble are intensifying. Andrew Chang explains ...

New



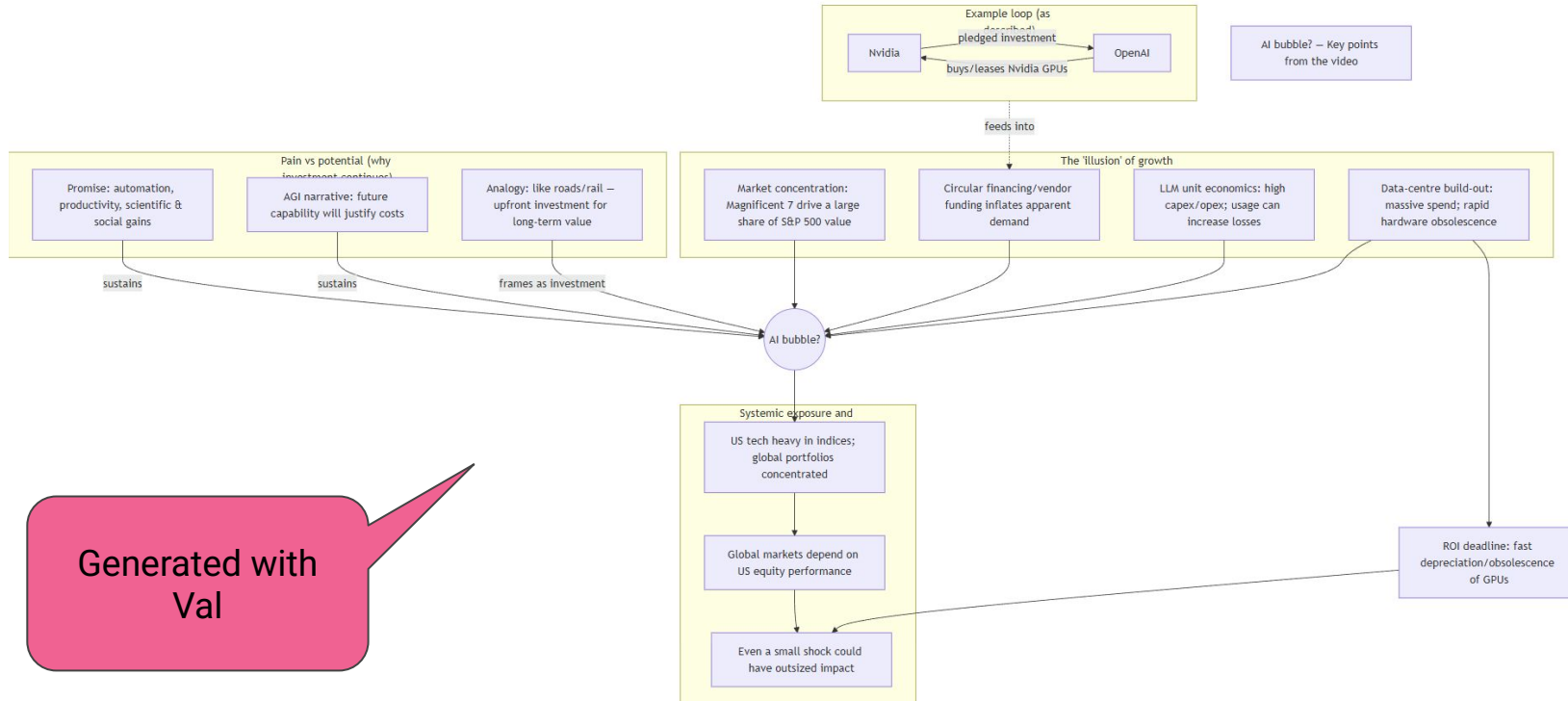
3 chapters Intro | The 'illusion' of growth | Pain vs. potential



Suggested Steps:

- In the video description, click “Show Transcript”
- Copy and paste the transcript to Val
- What insights you want to extract from the transcript?

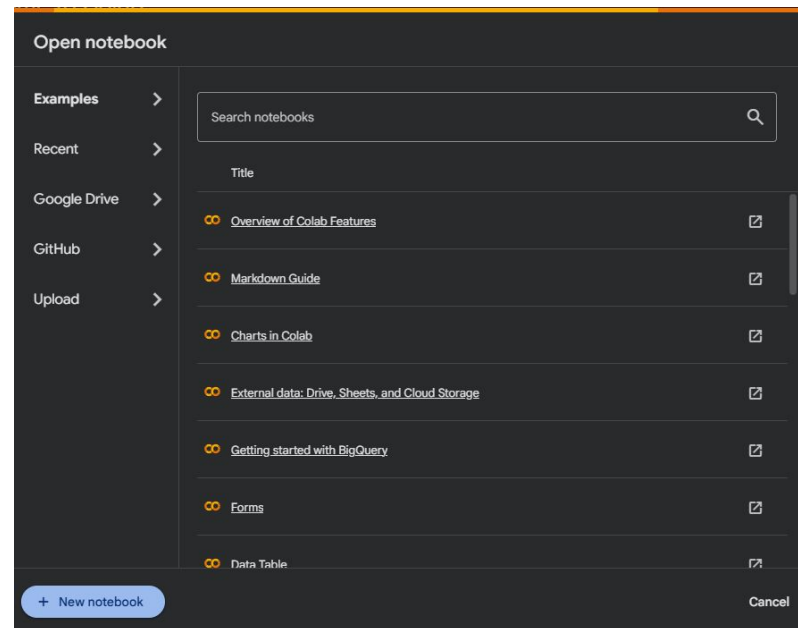
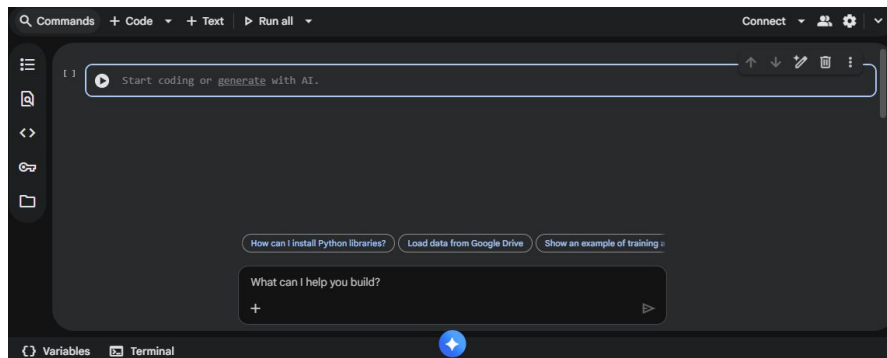
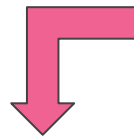
Visualisation of the video transcript



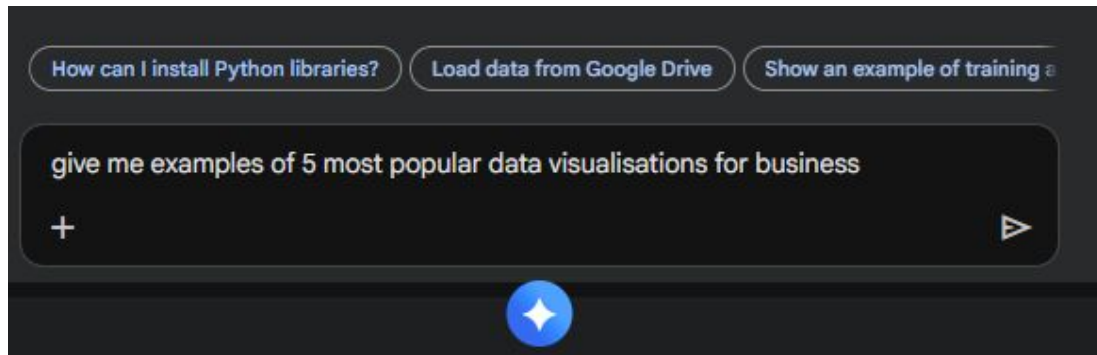
Generated with
Val

Walkthrough: Introducing Google Colab

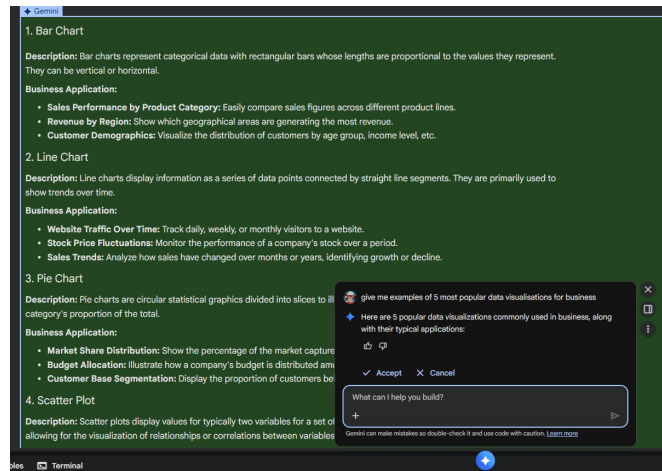
- Visit <https://colab.research.google.com/>
- Select New notebook



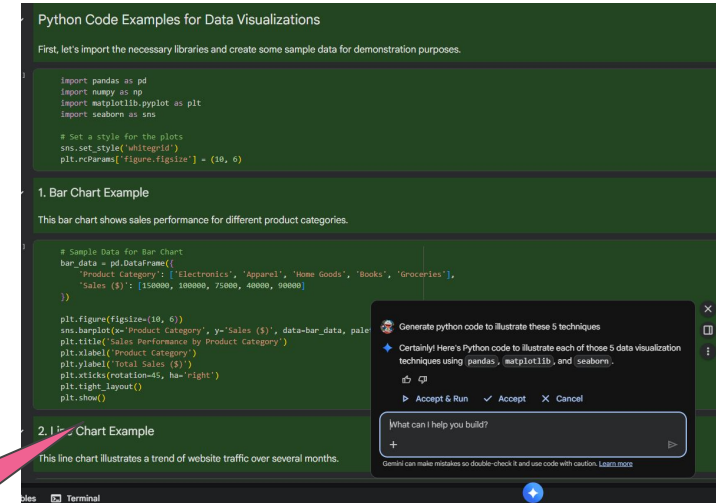
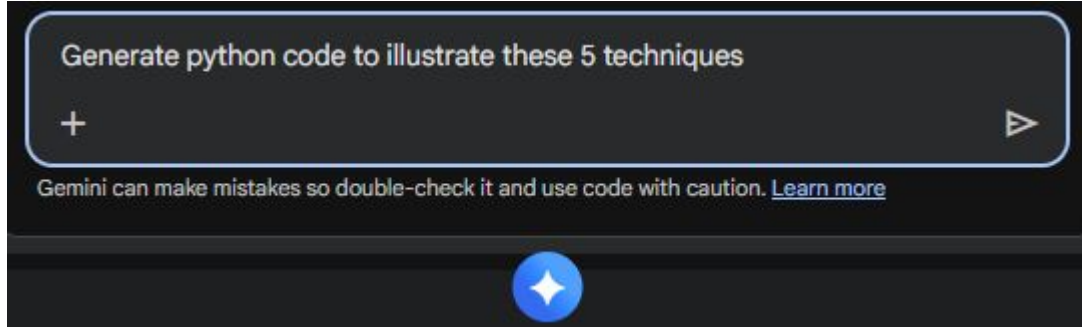
Try Gemini Agent in Google Colab



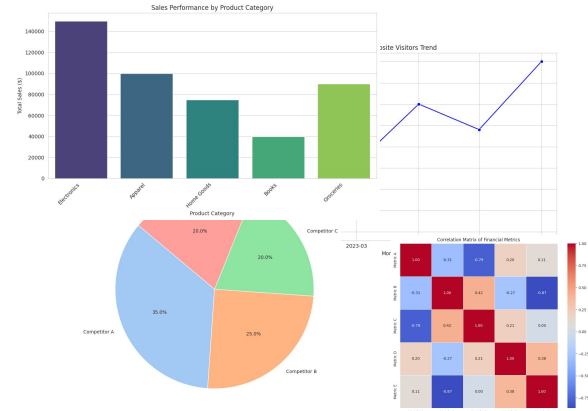
Behave like Val or ChatGPT, able to answer questions



Generate Python Code



Generate code which can be executed in Colab; just click **Accept and Run**



Demo: Use Google Colab and Val to analyse Melbourne CBD data

- Go to <https://data.melbourne.vic.gov.au/>
- Find the dataset for “City of Melbourne Dwellings and Household Forecasts by Small Area 2023-2043”
- Go to Export -> Download CSV file

City of Melbourne Dwellings and House

Information Table Analyze **Export** API

This dataset is licensed under : [CC BY](#)

Flat file formats

CSV [Whole dataset](#)

CSV uses comma (,) as a separator.

JSON [Whole dataset](#)

Excel [Whole dataset](#)

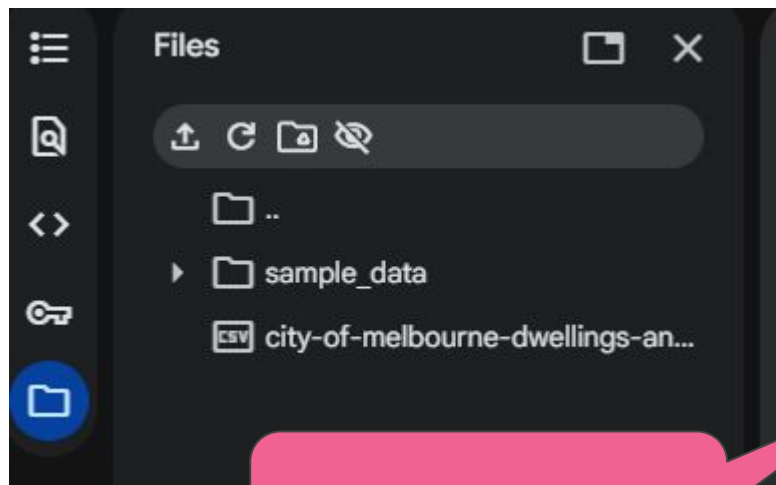
Data analysis file formats

Parquet [Whole dataset](#)

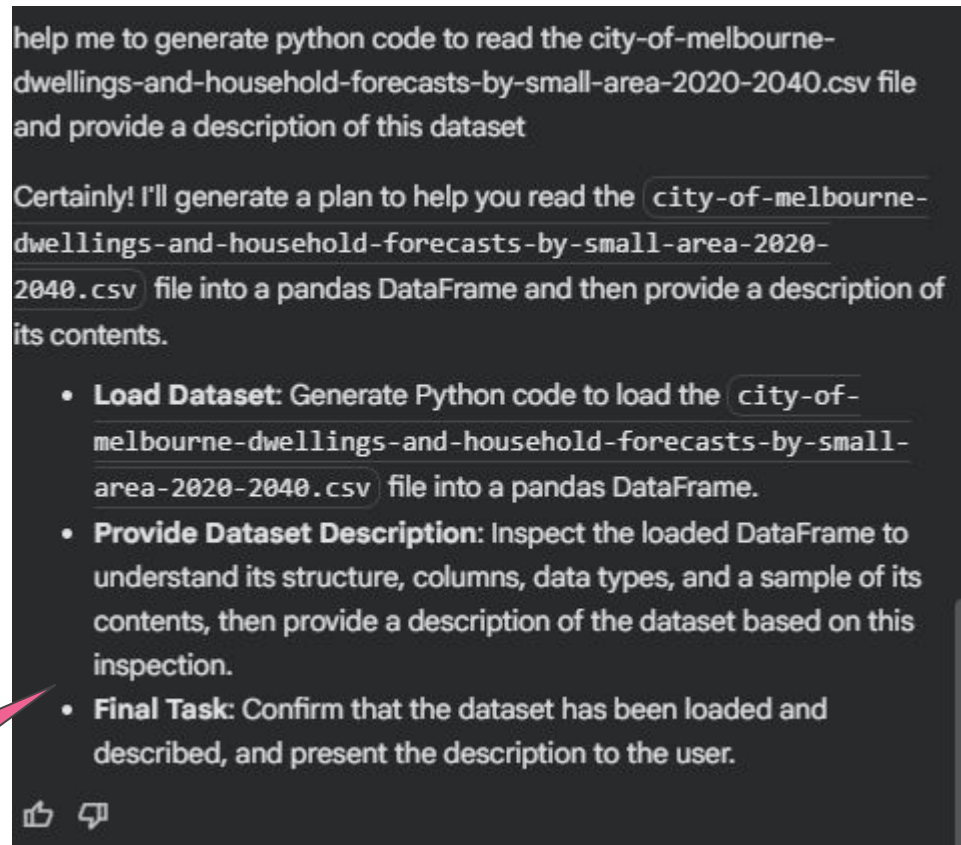
GeoParquet specification is supported for enhanced spatial data handling.

Upload Files to Colab

- Click on Folder icon
- Drag downloaded file to Files



Click **Accept and Auto-Run**



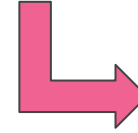
generate python code to visualise this datasets and help me gain further insights

Analyses

Dataset Description

The loaded dataset, `city-of-melbourne-dwellings-and-household-forecasts-by-small-area-2020-2040.csv`, has the following characteristics:

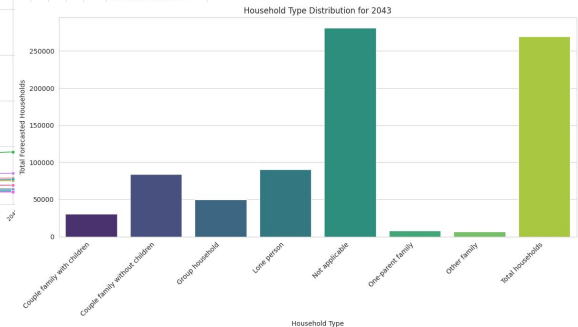
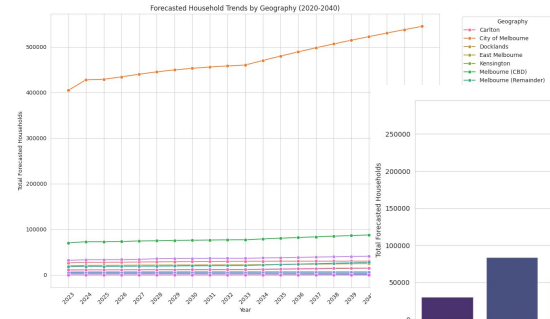
- **Overall Structure:** The DataFrame contains **2646 rows** and **5 columns**.
- **Column Details:**
 - `\uffeffGeography` :
 - **Data Type:** `object` (string).
 - **Non-null values:** 2646 (no missing values).
 - **Sample Content:** Contains geographical names like 'Carlton', 'Docklands', 'Melbourne (Remainder)', 'City of Melbourne'.
 - **Note:** The column name includes a Byte Order Mark (`\uffeff`), which indicates a potential encoding issue that might need to be addressed for cleaner column access.
 - `Year` :
 - **Data Type:** `int64` (integer).
 - **Non-null values:** 2646 (no missing values).
 - **Sample Content:** Represents forecast years (e.g., 2025, 2023, 2027, 2029, 2031, 2030).
 - `Category` :
 - **Data Type:** `object` (string).
 - **Non-null values:** 2646 (no missing values).
 - **Sample Content:** Describes the type of data, primarily 'Household type'.
 - `Households` :
 - **Data Type:** `object` (string).
 - **Non-null values:** 2646 (no missing values).
 - **Sample Content:** Categorizes household types, such as 'Other family', 'Couple family with children', 'Group household', 'Lone person', 'Couple family without children'.



✓ Preparing

✓ Clean Geography Column

- ⌚ Overall Forecasted Household Trend (2020-2040)
- ⌚ Forecasted Household Trends by Geography
- ⌚ Household Type Distribution (Latest Forecast Year)
- ⌚ Final Task



Gain insights of the case study (Assessment 1)

- Go to the simulator app
- Start Simulation
- When finished, download All Orders data
- Use Google Colab and the AI tricks demonstrated earlier to analyse this dataset

